

$$\frac{[Pr'_{Ce}](x)}{[Pr'_{Ce}]_{bulk}} = \exp\left(-\frac{ze}{kT}\Delta\phi(x)\right) \quad (2)$$

where the potential distribution  $\Delta\phi(x)$ , can be obtained by solving the Poisson equation to account for the total electroneutrality between the interface and the space charge layer<sup>12</sup>. An analytical solution is generally possible by making the depletion approximation, whereby the concentration of one of the carriers is negligible compared to the other (generally true when the two major carriers are of opposite sign, causing the depletion of one in the space charge region and the accumulation of the other).

In general, we can then define conductance parallel and perpendicular to the interface as:

$$\sigma_{//,sc} = \frac{1}{L_D} \int_0^{2L_D} [Pr^{3+}](x) \left(1 - \frac{[Pr^{3+}](x)}{[Pr_{tot}]}\right) q\mu_{Pr} dx \quad (3)$$

$$\rho_{\perp,sc} = \frac{1}{L_D} \int_0^{2L_D} \frac{1}{[Pr^{3+}](x) \left(1 - \frac{[Pr^{3+}](x)}{[Pr_{tot}]}\right) q\mu_{Pr}} dx \quad (4)$$

where  $L_D = \left(\frac{kT\varepsilon}{2e^2[Pr'_{Ce}]_{bulk}}\right)^{0.5}$  is the Debye length and  $\left(1 - \frac{[Pr^{3+}](x)}{[Pr_{tot}]}\right) = \frac{[Pr^{4+}](x)}{[Pr_{tot}]}$  reflects the fact that in PCO the mobility of localized electrons sitting on  $Pr^{3+}$  sites is only possible if adjacent available Pr sites are “empty” ( $Pr^{4+}$  state). The measured  $E_a$  in the space charge regions is thus expected to have multiple contributions, which include the sum of one-third of the reduction enthalpy of Pr in PCO, the small polaron hopping energy, and a contribution from the space charge potential. Directly solving these equations can become challenging especially in the scenario that  $\Delta\phi > 0$  where we can not simplify the site occupancy term  $\left(1 - \frac{[Pr^{3+}](x)}{[Pr_{tot}]}\right)$ . Moreover, one of the major assumptions in deriving the Gouy-Chapman model no longer holds, i.e., the assumption of a dilute limit condition. Nevertheless, for the sake of our study, we can visually inspect equations (3) and (4) to predict the expected behavior.

When  $\Delta\phi < 0$ , a situation described at the substrate/surface heterointerface of our nanowire in contact with an acidic oxide, then  $\frac{[Pr'_{Ce}](x)}{[Pr'_{Ce}]_{bulk}} < 1$  and  $[Pr'_{Ce}](x)$  becomes depleted in the vicinity of the interface. Conduction along  $\sigma_{//,sc}$  would systematically be bypassed by the parallel contribution of  $\sigma_{bulk}$  (**Supplementary Fig. 6a**).

When  $\Delta\phi > 0$ , a situation described at the substrate/surface heterointerface when our nanowire is in contact with a basic oxide or at the grain boundaries, then an accumulation of  $Pr^{3+}$  in the space charge region is expected to occur (**Supplementary Fig. 6b**). In the

accumulated space charge region, where  $[Pr^{3+}](x) \sim 50\%[Pr_{tot}]$ , then a local maximum in conductivity is expected, where  $[Pr^{3+}](x) \left(1 - \frac{[Pr^{3+}](x)}{[Pr_{tot}]}\right)$ , is maximized. Above this point, closer to the interface, the effective mobility of small polaron is expected to decrease with further increases in  $[Pr^{3+}](x)$ , owing to the decrease in site availability (i.e.  $[Pr^{3+}](x) \left(1 - \frac{[Pr^{3+}](x)}{[Pr_{tot}]}\right)$ ). The implication is that an anisotropic conductivity profile may arise, with a high resistivity region developing close to the interface wherever  $[Pr^{3+}](x) > 50\%[Pr_{tot}]$ , contributing to a series resistance for perpendicular transport across the interface, while a conductivity maxima where  $[Pr^{3+}](x) \sim 50\%[Pr_{tot}]$ , occurring a small distance away from the interface for parallel transport. The conductivity profile will follow the trends displayed in **Supplementary Fig. 7**:

The above rationale explains how electron accumulation at a heterointerface can enable enhanced conductivity parallel to the interface (for MgO and Li<sub>2</sub>O), while causing higher resistance for bulk transport across grain boundaries. It is also entirely consistent with the observations made previously by Zhang et al. in bulk nanocrystalline ceramic of PCO, when we consider that ceramic samples sintered at higher temperatures typically exhibit smaller grain boundary space charge potentials<sup>10</sup>. Nanocrystalline samples sintered at lower temperatures are therefore expected to have higher space charge potentials and therefore stronger Pr<sup>3+</sup> accumulation creating a more significant resistive blockage to perpendicular transport across the space charge region, even though the apparent activation energy is reduced.

Moreover, we recently showed that the GB space charge potentials in acceptor-doped CeO<sub>2</sub> thin films could be modulated through the in-diffusion of elements from the substrates at intermediate temperature (700-900 °C)<sup>13</sup>. We showed that depending on the element diffusing up to the GB (Al/Mg), increases or decreases in space charge potentials would be observed. This was enabled by the fact that cation diffusion along GB cores is accelerated relative to the bulk at these intermediate temperatures and the fact that Al<sup>3+</sup> ions are believed to sit on interstitial sites in GBs in ceria-based ceramics, inducing a net positive charge of 3+ ( $Al_i^{\bullet\bullet\bullet}$ ) and therefore an increase in space charge potential<sup>14-16</sup>, while Mg<sup>2+</sup> sits substitutionally, leading to a net negative charge of 2- ( $Mg_{Ce}''$ )<sup>13</sup>. The up-diffusion of Mg<sup>2+</sup> would be expected to decrease the space charge potential at the grain boundaries, while leading to a depletion of electrons at the substrate/surface interface. This grain boundary in-diffusion process would also explain why transport across grain boundaries will be more resistive than the bulk while

possessing a lower activation energy, as the sign of the space charge potential would actually act to reduce the total activation of conductivity composed of one-third of the reduction enthalpy and bulk mobility (note that in equation(1) and equation (2), the exponential terms have different signs). In the case of **Supplementary Fig. 8**, where the  $\text{PCO}_{\text{NA}}$  on  $\text{Al}_2\text{O}_3$  is annealed at  $700^\circ\text{C}$ ,  $\text{Al}^{3+}$  at the grain boundary is expected to cause a highly increased space charge potential and therefore a higher series resistance ( $\rho_{\perp}$ ) induced by the depressed small polaron mobility, contributing to a lower conductance. On the other hand at  $400^\circ\text{C}$ , the space charge potential at the grain boundaries is expected to be small and therefore the bulk conductivity of the film would align closer with bulk theoretical expectations. In **Supplementary Fig. 7**, we provide a 3D visualization of the anisotropic charge transport that would occur at surface and grain boundaries due to a  $\Delta\phi > 0$  and compare it to the situation of electron depletion at the substrate interface (i.e.  $\Delta\phi < 0$ ).